

Spinal reflexes

The written notes for the spinal reflexes walkthrough. Week 5 competencies 1 to 4 and 7.

The reflex arc

A reflex is an automatic, stereotyped response to a stimulus. Spinal reflexes are integrated in the spinal cord and do not need the brain.

1. Receptor: detects the stimulus (for the knee jerk, the muscle spindle).
2. Sensory (afferent) neuron: carries the signal into the cord through the dorsal root. Its cell body is in the dorsal root ganglion.
3. Integrating center: the spinal cord gray matter, where the sensory neuron synapses.
4. Motor (efferent) neuron: carries the response out of the cord through the ventral root.
5. Effector: the muscle or gland that responds (for the knee jerk, the quadriceps).

Monosynaptic reflex

The sensory neuron synapses directly on the motor neuron: one synapse. Fastest possible reflex. The stretch reflex is the example.

Polysynaptic reflex

One or more interneurons between the sensory and motor neuron. Slower, because each synapse adds a delay of roughly half a millisecond, but able to inhibit, to cross the midline, and to reach several segments.

Cut	Reflex	Sensation from that area	Voluntary movement
Dorsal root	Lost	Lost	Intact
Ventral root	Lost	Intact	Lost

Kinds of reflexes

Sorted by	Kinds	Example
Effector	Somatic motor (skeletal muscle) or autonomic (smooth muscle, cardiac muscle, glands)	Knee jerk; baroreceptor reflex
Integrating center	Spinal (spinal cord) or cranial (brain)	Knee jerk; pupillary light reflex (midbrain)
Learned or not	Innate (present from birth) or learned (acquired with experience)	Withdrawal reflex; salivating at the smell of a favorite food
Synapses	Monosynaptic or polysynaptic	Stretch reflex; withdrawal reflex

The pupillary light reflex is autonomic, cranial, innate and polysynaptic.

The stretch reflex

- The muscle spindle is a capsule of small intrafusal muscle fibers lying in parallel with the ordinary extrafusal fibers. Sensory endings wrap around the middle of the intrafusal fibers.
- Spindles report muscle length. They fire tonically even at rest, because the spindle is slightly stretched.
- Stretch the muscle: mechanically gated channels in the sensory endings open, the spindle afferent fires faster, and it excites the alpha motor neurons of the same muscle, which contracts. This is negative feedback on muscle length.
- Reciprocal inhibition: a branch of the spindle afferent excites an inhibitory interneuron, which inhibits the motor neurons of the antagonist, so the antagonist relaxes. Inhibition always needs an interneuron, because a sensory neuron's transmitter is excitatory.
- Alpha-gamma coactivation: gamma motor neurons contract the ends of the intrafusal fibers during a contraction, keeping the spindle taut so it keeps reporting length. Without them the spindle would go slack and silent.

Muscle spindle

Stretch receptor in parallel with the muscle fibers. Reports length.

Alpha motor neuron

Motor neuron to the extrafusal fibers that do the work.

Gamma motor neuron

Motor neuron to the ends of the intrafusal fibers of the spindle.

The Golgi tendon organ

- Collagen fibers at the junction of muscle and tendon with a sensory ending threaded between them. It lies in series with the muscle fibers.
- It reports tension (force). When the muscle pulls on the tendon, the collagen squeezes the ending and it fires.
- Its reflex inhibits the same muscle through an inhibitory interneuron, reducing force. It helps protect muscle and tendon from damaging loads and can make a muscle give way under an extreme load.

	Muscle spindle	Golgi tendon organ
Position	In parallel with the muscle fibers	In series, at the tendon
Reports	Length (stretch)	Tension (force)
Reflex effect on its own muscle	Excites: contract	Inhibits: relax
Synapses	Monosynaptic to its own muscle	Polysynaptic, through an inhibitory interneuron
Held still while the load rises	Steady firing	Firing rises

Withdrawal and crossed extensor reflexes

1. Nociceptors in the right foot fire (stepping on a tack).
 2. Sensory neurons enter the cord and excite interneurons.
 3. Same side: excitatory interneurons drive the flexor motor neurons, and inhibitory interneurons quiet the extensor motor neurons. The foot lifts. This is the flexion (withdrawal) reflex.
 4. Other side: interneurons cross the midline. The left extensors contract and the left flexors are inhibited, so the left leg takes the body's weight. This is the crossed extensor reflex.
 5. Interneurons also spread the signal up and down several segments, so the hip, knee and ankle move together.
- Both reflexes are polysynaptic: inhibition, crossing and spreading all need interneurons.
 - Without the crossed extensor half, the injured foot would still lift but the other leg would buckle under the load.

The brain's control of spinal reflexes

- Muscle tone: spindles fire at rest, keeping a few alpha motor neurons active, so a resting muscle is slightly firm.

Cutting the dorsal roots removes tone and the tendon reflex.

- Descending pathways from the brain act on spinal motor neurons, interneurons and gamma motor neurons, and on balance they hold spinal reflexes in check.
- Jendrassik maneuver: pulling the hooked hands apart during the tap is thought to raise descending drive and spindle sensitivity, so a weak reflex gets bigger. A reflex that reinforces has an intact arc.
- Plantar response: stroking the sole normally curls the toes down. A big toe that extends with the toes fanning is the Babinski sign, an upper motor neuron sign. It is normal in infants until about one to two years old, while the corticospinal tract matures.

	Upper motor neuron lesion	Lower motor neuron lesion
Where	Motor pathway in the brain or cord above the reflex (corticospinal tract)	Ventral horn motor neuron, ventral root or peripheral nerve
Strength	Weak	Weak
Tone	High (spasticity)	Low (flaccid)
Reflexes	Brisk, often with clonus	Reduced or absent
Plantar response	Babinski sign	Normal or absent
Over weeks	Little wasting	Muscle wasting and fasciculations

Testing reflexes

Grade	Meaning
0	Absent
1+	Less than usual
2+	Normal
3+	Brisker than usual
4+	Very brisk, often with clonus

- An absent reflex means the arc is broken somewhere: receptor, sensory neuron or dorsal root, cord segment, motor neuron or ventral root, or muscle (a lower motor neuron or sensory problem).
- A very brisk reflex with clonus usually means the descending pathways from the brain that normally hold reflexes in check are damaged (an upper motor neuron lesion).
- Compare left with right. A difference between sides matters more than one number.
- Latency separates a reflex from a voluntary response. The knee jerk appears a few tens of milliseconds after the tap. Catching a dropped ruler takes about 150 to 250

milliseconds, because the path runs through the eyes, the brain and down the cord, crossing many more synapses.